

amount of nickle by which the life of exhaust valve can be greatly prolonged.

The intake manifold used on engine is a "D" shape cross section which results in good atomizing and even distribution of the fuel mixture to each of the six cylinders.

The exhaust manifold is designed to reduce back pressure to a minimum. Located on the inside of the exhaust manifold is the thermostatically operated heat control valve.

This valve in the exhaust manifold directs the hot exhaust gases against the center of the intake manifold when the engine is cold as shown at the left of Figure 3. As the engine warms up, the heat control valve coil closes the valve and directs the exhaust gases away from the intake manifold as shown at the right of Figure 3. This thermostatic control results in the proper temperature of the incoming gases under all operating conditions.

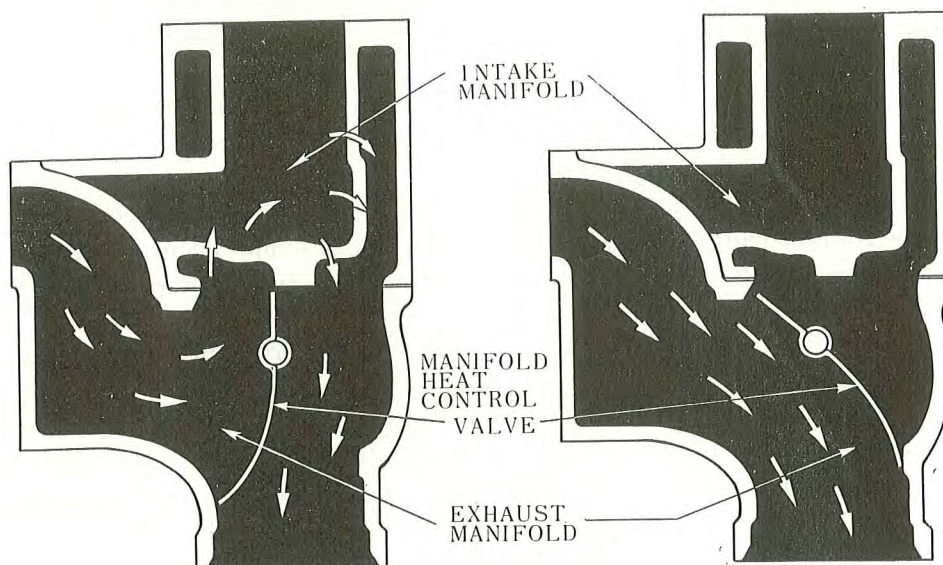


Figure 3—Manifold Heat Riser and Heat Control Valve

The tension of heat control valve coil is very important. When too tight, the heat will not be turned off the intake heat riser as the engine warms up, with the result that the incoming gases will be expanded several times greater than normal and it will be impossible to get a normal fuel charge into the cylinders. This condition will reduce power and maximum speed and cause detonation as well as sticking valves. Therefore, it is important that the heat control valve coil be wound up just enough to slip its outer end over the dowel pin (Fig. 4) in the manifold and no more. This is approximately $\frac{1}{2}$ turn of the coil from its position when unhooked.

Sometimes the heat control valve shaft becomes frozen in the manifold;